

ORIGINAL ARTICLE *Rare bleeding disorders*

The paradoxical association between inherited factor VII deficiency and venous thrombosis

S. MARTY,* C. BARRO,† B. CHATELAIN,‡ B. FIMBEL,§ B. TRIBOUT,¶ J. REYNAUD,** J.-F. SCHVED* and M. GIANZILY-BLAIZOT*

*CHU Montpellier, Laboratoire d'Hématologie, Hôpital Saint-Eloi, Montpellier, France; †CHU Grenoble, Département de Biologie Cellulaire, Grenoble, France; ‡UCL, Laboratoire d'Hématologie, Yvoir, Belgique; §CHU Tours, Laboratoire d'Hématologie, Tours, France; ¶CHU Amiens, Département de Chirurgie Vasculaire, Amiens, France; and **CHU Saint Etienne, Laboratoire d'Hématologie, Saint-Etienne, France

Summary. Inherited factor VII (FVII) deficiency is considered to be a haemorrhagic disease. Nonetheless, some patients paradoxically present with venous thrombosis. We assessed whether there was a link between phenotype and genotype in seven patients with inherited FVII deficiency and thrombosis (eleven venous thrombotic events). For each patient (FVII:C < 50%), clinical data were collected, aetiological assessment of risk factors for thrombosis was investigated, and direct sequencing of the nine exons and promoter of the *FVII* gene (*F7*) was performed. We present the second series ever published on FVII patients with thrombosis. In nine of the eleven thrombotic events, there was at least one classical triggering risk factor; clinical ($n = 4$), familial antecedent ($n = 2$), or biological, defined by phospholipid-binding antibodies or elevated FVIII:C levels ($n = 7$). In contrast to a previous series, only two events occurred after surgery, performed both with and without replacement therapy. The thrombotic

event remained unexplained in one young patient, highlighting the lack of 'protection' against venous thrombosis by low FVII:C levels. Genetic mutations were found to be heterogeneous. Among the seven *F7* sequence alterations identified in the present study, only two (p.Ala354Val and p.Arg364Gln) have previously been reported in FVII-deficient patients presenting with venous thrombosis. Our genetic analyses of the *F7* mutations in these patients show the complexity of FVII deficiency associated with thrombosis. These data justify a holistic, clinical and biological approach for patients with these specific symptoms. This series also strongly suggest that mild FVII deficiency should not prevent physicians from using antithrombotic prophylaxis in FVII-deficient patients.

Keywords: antithrombotic prophylaxis, *F7* mutations, inherited factor VII deficiency, thrombotic risk factors, venous thrombosis

Introduction

Inherited factor VII (FVII) deficiency is the most widespread of the rare inherited bleeding disorders, with an estimated prevalence of 1 in 400 000 Caucasians [1]. It is characterized by a wide heterogeneity as regards clinical, biological and genetic parameters. Clinical features are extremely variable,

ranging from mild cutaneo-mucosal bleeding to lethal cerebral haemorrhages, and are poorly correlated with residual FVII coagulant activity (FVII:C). Moreover, several patients remain asymptomatic [2], even under conditions of high haemorrhagic challenge. Notably, in some rare cases, patients have a history of arterial [3,4] or venous [5,6] thromboses. The mechanisms accounting for the association of FVII deficiencies with thrombosis remain unelucidated.

Here, we report a new series of seven patients with FVII deficiency having a history of at least one venous thrombosis (11 cases of thrombosis). The FVII genotype in each patient was characterized to determine whether a particular mutation and/or polymorphism of the *FVII* gene (*F7*) is associated with

Correspondence: Muriel Giansily-Blaizot, Laboratoire d'hématologie, CHU Montpellier, 80 avenue Augustin Fliche, 34295 Montpellier cedex 5, France.

Tel.: +33 467 33 7033; fax: +33 467 33 7036;
e-mail: m-giansily@chu-montpellier.fr

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these paradoxical cases. Additionally, thrombophilia markers were evaluated and their potential involvement in these cases is discussed.

Materials and methods

Patients

In a French series of 172 patients with inherited FVII deficiency, seven patients were identified as having suffered from a venous thrombosis. Inclusion criteria required a FVII:C level <50% of normal pooled plasmas and a previous history of at least one episode of venous thrombosis. Clinical features were recorded by the physician in charge of each patient in a standardized questionnaire focussed on FVII deficiency (circumstances of diagnosis, follow-up, treatments) and thrombotic events. Thrombophilia-screening included measurements of FVIII coagulant activity (FVIII:C), protein C (PC) activity, protein S (PS) activity, and antithrombin (AT) levels, and screening for the activated PC-resistance phenotype, factor V (FV) Leiden and prothrombin (PT) G20210A polymorphism, as well as the presence of lupus anticoagulant, anticardiolipin and anti- β 2GPI antibodies. All patients gave a written informed consent in accordance with local hospital ethics guidelines. None of the patients described here has been previously reported.

Characterization of biological FVII phenotypes

Measurements of FVII antigenic activity (FVII:Ag) and FVII:C levels from each hospital centre were checked in the St Eloi laboratory to assure standardization of results. FVII:C was measured according to a chrometric method based on the PT time principle, using a human recombinant thromboplastin (ORTHOPLASTIN™; Instrumentation-Laboratory, Lexington, MA, USA) and FVII-depleted plasma (Diagnostica Stago, Asnières-sur-Seine, France) as a substrate. For FVII:C levels of <10%, a sensitized technique was used [7]. For patients whose plasma was not available (patients nos 139, 143 and 151), levels determined by their respective centres are reported. FVII:Ag was measured according to an enzyme-linked immunosorbent assay (EIA) method using the ASSERACHROM™ FVII:Ag kit (Diagnostica Stago).

F7 genotyping

F7 genotyping was performed by direct sequencing according to previously published methods [8]. Each

sequence alteration was confirmed by enzymatic digestion, on at least two different PCR amplification products, when the altered sequence modified a restriction site or alternatively, by another independent sequence analysis. Mutations are designated in accordance with the standard international nomenclature guidelines recommended by the human genome variation society (HGVS at <http://www.hgvs.org/mutnomen/recs.html>), with nucleotide 1 being the A of the ATG-translation-initiation codon in the coding DNA. As amino acid number 1 refers to the translational initiation Met codon rather than the first amino acid in the mature FVII protein, the traditional names of mutations (based on previous nomenclature) are also provided in the brackets. The genomic (GenBank no. NC_000013.9) and mRNA (GenBank no. NM_000131) sequences of F7 were used as the reference sequences.

Results

Screening for thrombophilia

The age at which the first venous thrombosis occurred in our series of seven patients was very variable (median 44 years, range 16–65). An atypical localization of the thrombosis was found in four of the seven patients; portal and splenic veins, central retinal vein, axillary vein and cerebral veins. Three patients displayed bleeding diathesis, as documented by a mild tendency to easy bruising (patients nos 120 and 139) or a moderate postpartum haemorrhage (patient 119). Notably, patient 120 underwent several events involving a high risk of bleeding without any haemorrhagic complications. Aetiological assessment of thrombosis revealed the presence of known risk factors in six of the seven patients in our series. Three patients, who suffered five thrombotic events, exhibited an antiphospholipid syndrome. Patients 120 and 139 presented elevated FVIII:C albeit in the latter patient, high FVIII:C levels were associated with hyperfibrinogenaemia. Therefore, inflammation aetiologies could not be excluded. Four thrombotic events occurred subsequent to identifiable clinical triggering factors; postsurgical procedures ($n = 2$), prosthetic valve replacement ($n = 1$) and a lower limb compressive haematoma on account of a muscle tear ($n = 1$). Only one event occurred after substitution therapy with prothrombin complex-like concentrate (PCC). Notably, none of these seven patients had a deficiency of coagulation inhibitors, the FV Leiden mutation, or the 20210A allele. For patient 142, who neither smoked nor used oral contraceptives, no prothrombotic risk factor was identified (Table 1).

Table 1. Clinical features and *F7* genotypes of the seven FVII-deficient patients with venous thrombotic antecedents.

Patient_gender DB	Age (years) at the time of thrombosis	Thrombosis localization	Triggering factor	Substitutive therapy	Thrombosis prophylaxis	Familial thrombosis	Haemorrhagic symptoms	Haemorrhagic challenges without bleeding	APLS/ FVIII:C (%)	FVII:C/ FVII:Ag (%)	<i>F7</i> genotypes		
											Mutations	Allele-402A	Allele 353Q
57_M 1937	65	LL DVT	No	No	No	Daughter: DVT	No	No	No /<175	42/74	p.Gly391Ser/-	-	Heterozygous
119_F 1952	34	Central retinal vein	No	No	No	No	Moderate postpartum haemorrhage	Second delivery	Yes /<175	10/86	p.Arg364Gln/ p.Arg364Gln	Homozygous	-
	50	LL DVT + PE	No										
120_F 1925	57	LL DVT	Postsurgical (knee hygroma)	No	No	No	Easy bruising	Appendectomy ovariectomy traffic accident	Yes /217	9/73	p.Cys195Arg/ p.Arg364Gln	Heterozygous	-
	79	LL DVT + severe PE	No										
139_M 1959	44	LL DVT	LL compressive haematoma	No	No	Mother: DVT	Easy bruising	Unknown	No /290	50*/NA	p.Cys151Arg/-	-	-
	46	LL DVT	No										
142_F 1978	18	Upper limb DVT	No	No	No	No	No	No	No /<175	41/84	p.Ala354Val/ c.805 + 7A>G	-	Homozygous
143_F 1951	52	Mitral valve + cerebral venous thrombosis	Mitral prothetic valve	No	Oral anticoagulant therapy	No	No	No	No /<175	30_49*/ NA	c.805 + 7A>G/ c.805 + 7A>G	-	Homozygous
	52	LL DVT	Postsurgical (valve replacement)	PCC	No								
151_F 1989	16	Portal and splenic vein + Mesenteric infarction	No	No	No	No	No	No	Yes /<175	20*/NA	p.Gln160Arg/-	-	Heterozygous

The numbers in the first column refer to the patients' inclusion number in our French series of FVII deficiency. Thrombotic risk factors are highlighted in bold. **Patient 57:** his daughter presented a postsurgical LL DVP at the age of 18. **Patient 143:** her first thrombosis occurred upon mechanical mitral valve replacement in the context of atrial fibrillation, and the second occurred following surgery performed with substitution therapy.

DB, date of birth (year); DVT, deep venous thrombosis; F, female; LL, lower limb; M, male; NA, not available; PE, pulmonary embolism; PCC, prothrombin complex-like concentrates.

*Measured in the respective centre in charge with NEOPLASTINTM (Diagnostics Stago).

Characterization of FVII deficiencies

Four patients were classified as 'cross-reacting material positive' (CRM+) as they displayed decreased rates of FVII:C (Table 1) and normal antigen levels (74, 86, 73 and 84%, for patients nos 57, 119, 120 and 142, respectively). F7 genotyping revealed a variety of mutations, in agreement with the genotypic heterogeneity of inherited FVII deficiency. The p.Arg364Gln mutation (classically referred to as R304Q) as well as the c.805 + 7A>G sequence alteration (IVS7 + 7A>G) were each detected in three of the 14 alleles, while the p.Cys151Arg (C91R), p.Gln160Arg (Q100R), p.Cys195Arg (C135R), p.Ala354Val (A294V) and p.Gly391Ser (G331S) mutations were found each in one allele. Six of the 14 alleles harboured the p.413Gln (353Q) polymorphism, known to be associated with decreased FVII:C levels. Patients 119 and 120 were homozygous and heterozygous, respectively, for the c.-402A polymorphism (Table 1).

Discussion

In this study, we report clinical and genetic data for seven FVII-deficient patients displaying a paradoxical association with venous thrombosis. To our knowledge, this series is the second ever published on this subject, on account of the scarcity of this type of cases [9]. Notably, in our series, 10 of the 11 cases of venous thrombosis were independent of any clotting factor replacement therapy and only one was related to substitutive therapy.

Characteristics of the thrombotic events

FVII deficiency is characterized by a wide heterogeneity, even amongst those patients presenting with rare thrombotic events. In a few case reports, thrombosis can occur in 'usual' sites such as the deep veins of the lower limbs or pulmonary embolism, or in atypical sites such as the sinusal veins [10,11]. The first series of FVII deficiency associated with thrombosis included seven cases of venous thrombosis, localized primarily in typical sites (lower limbs and pulmonary embolism) [9]. In contrast to the previous report by Mariani *et al.*, our series outlines a relatively higher frequency of atypical sites of thrombosis. It is noteworthy that these atypical sites concerned younger patients (median 26 years, range 16–52). Both the early age of onset and the high percentage of atypical thrombotic sites may be indicative of a specific physiopathological process that remains to be defined. Indeed, with the excep-

tion of primary axillary venous thrombosis which is often associated with active use of the upper extremities or anatomic abnormalities in young healthy patients [12], atypical thrombotic sites remain anecdotal in children or young adults. In our series, retinal vein occlusion and portal venous thrombosis events were associated with antiphospholipid antibodies in patients 119 and 151, respectively, as previously reported in the literature [13,14]. On the other hand, the upper limb thrombosis appeared to be spontaneous as no mechanical aetiology was reported for patient 142. It is therefore important to evaluate more cases of FVII deficiency or FVII variants to determine whether there is a specific association between this disease and the specific venous thrombosis.

Our current study identifies only two postsurgical cases of thrombosis (one of which occurred concurrently with substitutive therapy). One possible explanation for the distinct 'causes' of thrombosis in our current study when compared with the previous series may be on account of differences between studied populations. In the Mariani series, five of the seven reported events occurred subsequent to the surgery, performed concurrently with substitutive therapy. Those patients who displayed lower FVII:C levels (median 3% vs. 41% in our series) were treated with replacement therapy from variety of sources (fresh frozen plasma, plasma-derived FVII and prothrombin-complex-like concentrates). In our series, patients displayed milder FVII deficiency, likely accounting for the fact that only one patient received PCC (KASKADIL™, LFB, France). Such PCCs adversely increase other vitamin K-dependent factors, which was found to be associated with adverse arterial and venous thrombotic events [9,15,16] usually identified at typical sites. Thus, although the surgery performed under substitutive therapy appears to be an incontestable triggering factor that is sufficient to provoke venous thrombosis in FVII-deficient patients, other risk factors also need to be considered. Indeed, in this series, 10 of the 11 events occurred independently of any clotting factor-substitutive therapy.

In four of these 10 thrombotic cases, clinical triggering factors of thrombosis or a familial thrombotic history could be identified. Previous reports of FVII deficiency associated with venous thrombosis did not precisely evaluate the biological data [11,17]. Here, we find that in the remaining six cases, five events could be associated with at least one biological risk factor for thrombosis; the presence of lupus anticoagulant, anticardiolipin and anti-β2GPI antibodies and/or elevated FVIII:C levels. Neither the FV

Leiden mutation nor the PT G20210A polymorphism nor abnormal levels of PC or PS were detected. Indeed, the presence of phospholipid-binding antibodies and elevated FVIII:C levels were the only biological risk factors revealed by our study. Elevated plasma levels of FVIII:C has recently been recognized as a strong risk factor for the occurrence of venous thromboembolism [18] and recurrence [19]. To our knowledge, this is the first time that this parameter has been evaluated in such paradoxical cases. Plasma levels of FVIII:C > 175% were observed and confirmed in two patients in our series. Unfortunately, FVIII:C plasma levels in relatives were not available for assessing whether this constitutes a familial trait. Lupus anticoagulants are well known risk factors for venous thrombosis [20] and were present in three of the seven patients, although they were negative in all previously reported cases [9]. As already reported for other inherited deficiencies of clotting factors, coinheritance of thrombotic risk factors may modify the clinical phenotype. It has been well documented for the FV Leiden [21,22], the PT20210A polymorphism [23] and the FV HR2 haplotype [24], but it may also be relevant in other pro-coagulant contexts such as antiphospholipid syndrome.

To summarize, in most of our cases (10 of the 11 thrombotic events), there was at least one classical triggering risk factor either clinical or biological, contributing to a hypercoagulability state. In the remaining case (patient 142), no risk factor for thrombosis was identified, raising the question of whether her FVII genotype could explain that thrombotic phenotype. The ensemble of our data indicates that FVII deficiency does not offer protection when a certain threshold of hypercoagulability has been reached. Consequently, mild inherited FVII deficiency should not prevent physicians from using perioperative antithrombotic prophylaxis for venous thromboembolism.

F7 genotypes and thrombosis

To evaluate a possible link between FVII deficiency and thrombosis, we assessed whether an abnormality in the *F7* results in a tendency towards hypercoagulability. Molecular analysis of the mutated *F7* gene revealed a wide spectrum of *F7* genotypes. First, the presence of both p.Gln160Arg (Q100R) and p.Cys195Arg (C135R) mutations in our series was surprising. These mutations, in the homozygous state, have been found to be associated with severe haemorrhagic diathesis [25]. The p.Gln160Arg mutation has been reported to result in a major

secretory defect attributable to a misfolding mutant molecule [26]. The p.Cys195Arg mutation is assumed to significantly affect the function, secretion and stability of the FVII molecule, because it suppresses a single disulphide bridge between both chains of FVIIa [25,27]. Likewise, the newly described p.Cys151Arg (C91R) missense mutation results in the loss of the disulphide bond between Cys151 and Cys162 in the EGF2 domain. The mutant FVII protein is predicted to be misfolded and as such, to result in an impaired secretion of the protein. These mutations, present in a heterozygous state in our series, are therefore expected to result in significantly decreased FVII:C and FVII:Ag levels (<50%). Thus, these data highlight the fact, that low levels of FVII:C do not provide protection from thrombosis.

Notably, the four other mutations in our series are classically associated with asymptomatic presentations. The question of whether the c.805 + 7A>G sequence alteration should be classified as a mutation or a polymorphism remains open [28,29]. Similarly, the p.Gly391Ser mutation was reported to be compatible with a normal initiation of the coagulation cascade. Patients carrying this mutation have normal plasma FVII:Ag levels. They are asymptomatic or present with only moderate haemorrhagic clinical symptoms. *In vitro*, recombinant or purified plasmatic p.Gly391Ser FVII variants are able to generate rates of FXa [30]. Although, this is lower than that found in normal plasma, the residual activity of this FVII variant would be sufficient to initiate coagulation. Therefore, the p.Gly391Ser mutation is not incongruous with respect to the thrombotic phenotype.

In addition to the apparent heterogeneity in *F7* mutations, it is interesting to note that the p.Ala354Val and p.Arg364Gln mutations have already been reported to be associated with venous thrombosis [6,9]. The p.Ala354Val mutation was over-represented in a previous series of seven venous thrombotic episodes (five of the 14 studied alleles) as its frequency in Caucasian FVII deficiency is significantly lower [9]. In addition, the p.Arg364Gln mutation, identified in three of 14 alleles (allelic frequency of 21%) in our study, has been previously reported in an FVII-deficient patient presenting with a pulmonary embolism [6]. Both these mutations have been studied *in vitro*. A purified p.Arg364Gln FVII molecule was reported to have an impaired TF-binding, when compared with the wild type [31]. Results of the investigations of the p.Ala354Val mutation have been contradictory, showing either a lower [32] or similar [33] affinity for TF compared

with the wild type. This latter feature could be related to our *in vivo* observation of a thrombotic event in a young female patient without any known risk factor who carries the p.Ala354Val mutation.

Conclusions

This study of seven patients with inherited FVII deficiency and concomitant venous thrombosis underlines the importance of the aetiological assessment of thrombosis. It demonstrates again that low FVII:C levels do not protect against thrombosis, thereby indicating the relevance of perioperative thrombosis prophylaxis for these patients. For mild FVII deficiencies (FVII levels of 30–50% of normal) wherein patients are asymptomatic, physicians should not hesitate to undertake antithrombotic prophylaxis even in the absence of any FVII replacement therapy. For less mild FVII deficiencies, the prophylaxis for thrombosis may be associated with a moderate substitution therapy [9,34]. Moreover, patients undergoing substitution therapy alone should be carefully evaluated with regards to the risk of thrombosis [35]. Finally, the question of whether some F7 genotypes are linked to a thrombotic phenotype remains unanswered and will require the evaluation of a larger series of FVII-deficient patients with venous thrombosis.

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Disclosures

The authors stated that they had no interests which might be perceived as posing a conflict or bias.

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